

DSAA3073: Theories in Data Science

Week 3: Agnostic PAC Learning, Uniform Convergence & Fundamental Theorem

Concise Learning Sheet

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Explorative projection

Opening Challenge

The Agnostic Learner's Dilemma

Last week we assumed the target concept $c \in \mathcal{H}$ —the realizable case. But what if this assumption fails?

- Your training data is noisy: some labels are wrong
- Your hypothesis class is too simple: no $h \in \mathcal{H}$ perfectly fits the data
- You don't know which scenario you're in

Can we still learn? What guarantees can we provide? This week we tackle the **agnostic** setting where we make no assumptions about realizability, and discover the **Fundamental Theorem of Statistical Learning** that characterizes exactly when learning is possible.

Introduction

Last week we studied PAC learning in the **realizable** case, where we assumed there exists a perfect hypothesis $h^* \in \mathcal{H}$ with zero error. This week we remove this assumption and enter the **agnostic** setting—the real world where no perfect hypothesis may exist. The formal definition and full implications are developed in the next section.

Learning objectives: By the end of this week, you will:

1. Formalize agnostic PAC learning and understand **uniform convergence** (simultaneous convergence of empirical error to true error for all hypotheses)
2. Prove generalization bounds using **symmetrization techniques** (replacing unknown expectations with “ghost sample” empirical estimates)
3. Apply the Fundamental Theorem of Statistical Learning
4. Understand the **No Free Lunch theorem** (why unrestricted hypothesis classes cannot be learned) and its implications
5. Connect sample complexity to hypothesis class complexity

The learning journey this week: We’ll start by formalizing the agnostic setting, then discover that ERM works if and only if **uniform convergence** holds. This leads us to ask: when does uniform convergence hold? The answer—finite VC dimension—is the Fundamental Theorem. We’ll prove this using **symmetrization**, derive concrete sample complexity bounds, and see that restrictions on $\mathcal{H}$ are **necessary** (No Free Lunch). Finally, we’ll see how to choose among multiple hypothesis classes (SRM).

Every concept builds on the last, forming a complete theory of when and how learning is possible.

Connection

From Week 2: We build on VC dimension from last week. Recall that $\text{VCdim}(\mathcal{H})$ measures the capacity of $\mathcal{H}$ to shatter sets. This week, VC dimension becomes the key to understanding when agnostic learning is possible—see the Fundamental Theorem in Section 5.

Example: Recurring scenario: Medical diagnosis

Throughout this week, we’ll follow this example: You’re building a classifier to predict disease from symptoms using linear classifiers. The true relationship is nonlinear, so even the best linear classifier has 15% error. With 1000 noisy training examples, can you find a linear classifier with 17% error? Watch how each concept we develop applies to this scenario.

The Agnostic PAC Learning Framework

Motivation: Why Agnostic?

Thought Experiment

The Noisy World

You're building a spam classifier. Your training data comes from users who sometimes mislabel emails:

- Legitimate emails marked as spam (false positives)
- Spam emails marked as legitimate (false negatives)
- Inconsistent labeling: same email, different labels

Even if you had infinite data, no classifier could achieve zero error. The best you can do is match the performance of the optimal classifier in your hypothesis class.

Definition: Agnostic PAC Learning

A hypothesis class $\mathcal{H}$ is **agnostic PAC-learnable** if there exists an algorithm A and a sample-complexity function $m_{\mathcal{H}}(\varepsilon, \delta)$ (depending on error tolerance ε , confidence δ , and the complexity of $\mathcal{H}$) such that:

For all $\varepsilon > 0$, $\delta > 0$, and all distributions $\mathcal{D}$ over $\mathcal{X} \times \mathcal{Y}$ (joint distribution over inputs and labels):

If $m \geq m_{\mathcal{H}}(\varepsilon, \delta)$, then with probability at least $1 - \delta$ over $S \sim \mathcal{D}^m$ (the random training sample):

$$R(A(S)) \leq \inf_{h \in \mathcal{H}} R(h) + \varepsilon \quad (1)$$

where $R(h) = \mathbb{P}_{(x,y) \sim \mathcal{D}}[h(x) \neq y]$ is the **true risk** (probability of misclassification). For finite-VC classes, $m_{\mathcal{H}}$ is controlled by $\text{VCdim}(\mathcal{H})$.

In words: The algorithm finds a hypothesis whose error is at most ε worse than the best possible hypothesis in $\mathcal{H}$. We're competing with the best in class, not trying to achieve zero error.

Technical note: If the infimum is not attained (no best hypothesis exists in $\mathcal{H}$), we use approximate ERM: find h_S with $\hat{R}_S(h_S) \leq \inf_{h \in \mathcal{H}} \hat{R}_S(h) + \frac{\varepsilon}{2}$. The analysis then gives $R(h_S) \leq \inf_{h \in \mathcal{H}} R(h) + \varepsilon$ as required.

Key differences from realizable case:

- **No assumption** that $\min_{h \in \mathcal{H}} R(h) = 0$ (there may be no perfect hypothesis)
- **Goal:** compete with best hypothesis in $\mathcal{H}$ (achieve $R(h) \leq \min_{h \in \mathcal{H}} R(h) + \varepsilon$), not achieve zero error
- **More realistic** but potentially harder (requires more samples)

Pause and Think

Question: In the realizable case, we required $R(A(S)) \leq \varepsilon$. In the agnostic case, we require $R(A(S)) \leq \min_{h \in \mathcal{H}} R(h) + \varepsilon$. Why is this weaker?

Hint: Consider what happens when $\min_{h \in \mathcal{H}} R(h) > 0$.

Answer:

If the best hypothesis has error 0.3, the agnostic guarantee allows us to have error up to $0.3 + \varepsilon$, while the realizable guarantee would require error $\leq \varepsilon$. The agnostic setting acknowledges we can't beat the best in class.

✓ Checkpoint

Quick check: In the agnostic setting with $\min_{h \in \mathcal{H}} R(h) = 0.2$ and $\varepsilon = 0.05$, what error rate does PAC learning guarantee?

Answer: At most $0.2 + 0.05 = 0.25$ with probability $\geq 1 - \delta$. We compete with the best in class (0.2) plus tolerance ε .

Empirical Risk Minimization (ERM)

Empirical Risk Minimization (ERM) is the natural learning algorithm: choose the hypothesis that performs best on the training data.

Definition: Empirical Risk Minimization

Given training sample $S = \{(x_1, y_1), \dots, (x_m, y_m)\}$, the **ERM algorithm** returns:

$$h_S = \operatorname{argmin}_{h \in \mathcal{H}} \hat{R}_S(h) = \operatorname{argmin}_{h \in \mathcal{H}} \frac{1}{m} \sum_{i=1}^m \mathbb{1}_{h(x_i) \neq y_i} \quad (2)$$

where $\hat{R}_S(h)$ is the **empirical risk** (fraction of training examples misclassified by h , defined in Week 2).

In words: ERM selects the hypothesis that makes the fewest mistakes on the training sample.

Intuition: If we can't identify the true best hypothesis (since we don't know the true distribution $\mathcal{D}$), at least find the one that performs best on our sample.

Wrong Path: Why Not Just Minimize Training Error?

“ERM always returns the hypothesis with lowest training error. Isn’t that enough?”

Why this fails: Overfitting! A complex hypothesis class can memorize the training data but fail on new data.

Example: Nearest neighbor classifier achieves $\hat{R}_S(h) = 0$ on any dataset but may have terrible generalization.

The key question: When does ERM generalize? When is $R(h_S) \approx \min_{h \in \mathcal{H}} R(h)$?

Connection

Preview of uniform convergence: The answer is **uniform convergence**. We need $\hat{R}_S(h)$ to approximate $R(h)$ uniformly for **all** $h \in \mathcal{H}$ simultaneously. This is the bridge between training and test performance, and it’s controlled by VC dimension.

Uniform Convergence

But wait—how do we **know** that ERM actually works? We defined ERM as the algorithm that minimizes training error, but we need $R(h_S) \approx \min_{h \in \mathcal{H}} R(h)$ to hold. What mathematical property guarantees this?

The answer is **uniform convergence**. This is the bridge between what we observe (training error) and what we care about (test error). Let's make this precise.

week: 3, title: "Section 2.1: Why Uniform Convergence? (Regenerated)")

Uniform Convergence: Why Guarantees for All Hypotheses?

But wait—how do we **know** that ERM actually works? We defined ERM as the algorithm that minimizes training error, but we need $R(h_S) \approx \min_{h \in \mathcal{H}} R(h)$ to hold. What mathematical property guarantees this?

The answer is **uniform convergence**. But before we define it formally, let's understand **why** we need such a strong property.

The Question Students Ask

Student Question

“We only output one hypothesis h_S after training. Why do we need the empirical error to converge to true error for **EVERY** hypothesis in $\mathcal{H}$? Isn't it enough that $\hat{R}_S(h_S) \approx R(h_S)$ for just the one we picked?”

This seems like a reasonable objection. After all:

- We train on sample S and output a single hypothesis h_S
- We use h_S for all future predictions
- We never touch the other hypotheses in $\mathcal{H}$

So why should we care about convergence for hypotheses we'll never use?

The Natural But Wrong Approach

You might think: “Let's apply Hoeffding's inequality to h_S after we select it.”

This approach seems perfectly reasonable. For any **fixed** hypothesis h , we know the empirical risk is:

$$\hat{R}_S(h) = \frac{1}{m} \sum_{i=1}^m \mathbb{1}_{h(x_i) \neq y_i} \quad (3)$$

Each indicator is an i.i.d. random variable in $[0, 1]$ with expectation $R(h)$. By Hoeffding's inequality:

$$\mathbb{P}[|\hat{R}_S(h) - R(h)| > \varepsilon] \leq 2 \exp(-2m\varepsilon^2) \quad (4)$$

Setting this $\leq \delta$ and solving for m :

$$m \geq \frac{1}{2\varepsilon^2} \ln\left(\frac{2}{\delta}\right) \quad (5)$$

Concrete calculation: With $\varepsilon = 0.1$ and $\delta = 0.05$:

$$m \geq \frac{1}{2 \cdot 0.01} \ln(40) \approx 50 \cdot 3.69 \approx 185 \text{ samples} \quad (6)$$

Only 185 samples! That seems very efficient. Problem solved, right?

Pause and Think

Question: What could possibly go wrong with this approach?

Hint: Think about **when** we chose h_S relative to **when** we observed the sample S .

Answer:

We chose h_S **after** seeing S , specifically because it had low $\hat{R}_S(h_S)$ on that sample!

Where This Breaks: The Selection Bias Trap

⚠ Crisis

The Fatal Flaw

The Hoeffding bound is only valid if we chose h **before** seeing the data.

But we chose h_S **after** seeing S —specifically because it minimized $\hat{R}_S(h)$ on that sample!

This creates **selection bias**: we're using the same data to both select h_S and estimate its error.

Let's see why this is catastrophic with concrete numbers.

The Selection Bias Problem

Suppose $|\mathcal{H}| = 1000$ hypotheses and $m = 185$ samples (from our calculation above).

For each individual hypothesis h , Hoeffding tells us:

$$\mathbb{P}[|R(h) - \hat{R}_S(h)| > 0.1] \leq 2 \exp(-2 \cdot 185 \cdot 0.01) = 2 \exp(-3.7) \approx 0.049 \quad (7)$$

So there's roughly a 5% chance that any specific hypothesis h has $|R(h) - \hat{R}_S(h)| > 0.1$.

But ERM doesn't pick a specific hypothesis! It picks:

$$h_S = \operatorname{argmin}_{h \in \mathcal{H}} \hat{R}_S(h) \quad (8)$$

What's the probability that **at least one** hypothesis in $\mathcal{H}$ has a large deviation?

$$\begin{aligned}
& \mathbb{P}[\exists h \in \mathcal{H} : |R(h) - \hat{R}_S(h)| > 0.1] \\
& \approx 1 - (1 - 0.049)^{1000} \\
& \approx 1 - (0.951)^{1000} \\
& \approx 1 - e^{-50} \\
& \approx 0.9999...
\end{aligned}$$

Notice the gap: the naive approach suggests $m = 185$ is enough, but accounting for selection bias requires $m \approx 11000$ samples—nearly **60 times more!**

Almost certainly, at least one hypothesis will have $\hat{R}_S(h)$ much lower than $R(h)$ by random luck!

ERM specifically selects such “unlucky” hypotheses—those with artificially low training error.

Numerical Demonstration

Let’s see what happens with different scenarios:

Scenario	Method	Sample bound	Valid?
Fixed h chosen beforehand	Hoeffding directly	$m = 185$	✓ Yes
ERM h_S chosen from data	Hoeffding on h_S	$m = 185$	✗ Selection bias!
ERM from $ \mathcal{H} = \infty$	Union bound	$m = \infty$	✗ Fails completely
ERM with $d = 11$ (VC dimension)	Uniform convergence	$m \approx 11000$	✓ Yes

The Trap Visualized

Here’s what happens when we use ERM with $m = 185$ samples:

1. We draw sample S of size 185
2. By random chance, several hypotheses in $\mathcal{H}$ will have $\hat{R}_S(h) < R(h)$ (lucky on this sample)
3. ERM picks $h_S = \operatorname{argmin}_h \hat{R}_S(h)$ —very likely one of these “lucky” hypotheses
4. We think $R(h_S) \approx \hat{R}_S(h_S)$ (low error!)
5. But actually $R(h_S) \gg \hat{R}_S(h_S)$ (much higher test error)
6. **Result:** Overfitting! Our model looks great on training data but fails on test data.

Wrong Path: Why Not Test on Held-Out Data?

“Can’t we just split our data and test h_S on a held-out set?”

Why this is incomplete: Yes, you can (and should!) validate on held-out data. But:

- This doesn’t tell us **how much** data we need initially
- We need **theory** to predict sample complexity before collecting data
- Uniform convergence provides the theoretical foundation

The practical split (train/validation/test) is a consequence of the theory, not a replacement for it.

The Key Insight

💡 Aha Moment

The Resolution

We need **uniform convergence**: guarantee that $|R(h) - \hat{R}_S(h)| \leq \varepsilon$ simultaneously for **ALL** $h \in \mathcal{H}$ with high probability.

Then no matter which h_S the algorithm picks—even in a data-dependent way—the guarantee holds!

This eliminates selection bias: the bound holds for **every** hypothesis simultaneously, including the data-dependent choice $h_S = \operatorname{argmin}_h \hat{R}_S(h)$.

Key principle: Control the worst case over all hypotheses, and you automatically control any specific choice (including data-dependent choices).

Why This Works

If we can guarantee:

$$\mathbb{P}_S \left[\sup_{h \in \mathcal{H}} |R(h) - \hat{R}_S(h)| \leq \varepsilon \right] \geq 1 - \delta \quad (10)$$

Then with probability $\geq 1 - \delta$, the following holds **simultaneously** for all $h \in \mathcal{H}$:

$$|R(h) - \hat{R}_S(h)| \leq \varepsilon \quad (11)$$

In particular, it holds for $h_S = \operatorname{argmin}_h \hat{R}_S(h)$, regardless of how we chose it!

The Formal Solution

Now we can formally define uniform convergence:

Definition: Uniform Convergence

For i.i.d. samples and 0-1 loss in this section, we say that **uniform convergence** holds for hypothesis class $\mathcal{H}$ if:

$$\lim_{m \rightarrow \infty} \mathbb{P}_S \left[\sup_{h \in \mathcal{H}} |R(h) - \hat{R}_S(h)| > \varepsilon \right] = 0 \quad (12)$$

for all $\varepsilon > 0$ and all distributions $\mathcal{D}$ over $\mathcal{X} \times \mathcal{Y}$.

In words: As the sample size m grows, the probability that **any** hypothesis in $\mathcal{H}$ has empirical error far from its true error goes to zero. The convergence is **uniform**—it happens for all hypotheses simultaneously at the same rate.

Here $\sup_{h \in \mathcal{H}}$ denotes the **supremum** (least upper bound)—the worst-case deviation over all hypotheses. The key requirement: this worst-case deviation becomes small with high probability.

Finite-sample form: For any $\varepsilon, \delta > 0$, if $m \geq m_0(\varepsilon, \delta)$, then with probability at least $1 - \delta$:

$$\sup_{h \in \mathcal{H}} |R(h) - \hat{R}_S(h)| \leq \varepsilon \quad (13)$$

Connection

Why “uniform”? We need the bound to hold for **all** hypotheses simultaneously, not just for a fixed hypothesis chosen beforehand. This is much stronger than pointwise convergence and is essential for ERM to work with data-dependent hypothesis selection.

Interpretation: The Bridge to Generalization

With enough samples, empirical risk $\hat{R}_S(h)$ approximates true risk $R(h)$ **uniformly** for all hypotheses in $\mathcal{H}$ simultaneously (not just for individual fixed hypotheses).

This is the bridge between:

- **What we observe:** Training error $\hat{R}_S(h)$ on sample S
- **What we care about:** Test error $R(h)$ on future data

Why This Solves the ERM Problem

🎯 Key Takeaway

Uniform convergence is the bridge between training performance and test performance. If it holds, and if the relevant infima are attained (or if we use approximate ERM), then:

$$R(h_S) \leq \hat{R}_S(h_S) + \varepsilon \leq \hat{R}_S(h^*) + \varepsilon \leq R(h^*) + 2\varepsilon \quad (14)$$

where $h^* = \operatorname{argmin}_{h \in \mathcal{H}} R(h)$ and $h_S = \operatorname{argmin}_{h \in \mathcal{H}} \hat{R}_S(h)$.

In words: ERM's test error is at most 2ε worse than the best possible hypothesis in $\mathcal{H}$.

Proof. Let $h^* = \operatorname{argmin}_{h \in \mathcal{H}} R(h)$ and $h_S = \operatorname{argmin}_{h \in \mathcal{H}} \hat{R}_S(h)$.

By uniform convergence, with probability at least $1 - \delta$:

$$\forall h \in \mathcal{H} : |R(h) - \hat{R}_S(h)| \leq \varepsilon \quad (15)$$

Therefore:

$$\begin{aligned} R(h_S) &\leq \hat{R}_S(h_S) + \varepsilon && \text{(uniform convergence for } h_S) \\ &\leq \hat{R}_S(h^*) + \varepsilon && \text{(ERM minimizes } \text{Remp} \text{)} \\ &\leq R(h^*) + 2\varepsilon && \text{(uniform convergence for } h^*) \end{aligned} \quad (16)$$

The first inequality uses uniform convergence for h_S (even though h_S was chosen based on the data).

The second inequality uses the defining property of ERM: h_S minimizes empirical risk, so $\hat{R}_S(h_S) \leq \hat{R}_S(h^*)$.

The third inequality uses uniform convergence for h^* : $\hat{R}_S(h^*) \leq R(h^*) + \varepsilon$. ■

The Convergence Spectrum

To fully appreciate uniform convergence, let's contrast it with weaker notions:

🧠 Mental Model: Uniform vs. Pointwise Convergence

Pointwise: For each fixed h , as $m \rightarrow \infty$, $\hat{R}_S(h) \rightarrow R(h)$.

- Each hypothesis converges on its own timeline
- Different hypotheses may require different sample sizes
- Useless for ERM (which picks h based on data!)

Uniform: For all $h \in \mathcal{H}$ simultaneously, $\sup_{h \in \mathcal{H}} |R(h) - \hat{R}_S(h)| \rightarrow 0$.

- All hypotheses converge together
- Single sample size works for entire class
- Rate independent of which h we pick
- Essential for data-dependent selection (ERM)

Analogy: Pointwise is like runners finishing a race at different times. Uniform is like all runners finishing within 1 second of each other—the pack stays together.

✓ Checkpoint

Quick check: Why can't we just apply concentration inequalities to a single fixed hypothesis h to justify ERM?

Answer: Because ERM chooses h_S based on the data—it's data-dependent. We need bounds that hold uniformly for all $h \in \mathcal{H}$, including the one ERM selects. Applying Hoeffding to h_S after selection creates selection bias.

Looking Ahead

Progress on backbone: Uniform convergence is the condition that makes ERM work. But this raises a new question:

🧐 Looking Ahead

When does uniform convergence hold?

We've established that uniform convergence is necessary for ERM to succeed. But:

- For which hypothesis classes $\mathcal{H}$ does it hold?
- How many samples m are required?
- Is there a simple condition that characterizes it?

The answer—revealed in the Fundamental Theorem—is **finite VC dimension**. This is the key that unlocks everything.

Finite Hypothesis Sets: Agnostic Bound

Before we tackle the general case, let's see how uniform convergence works for finite hypothesis classes. This will build intuition for the general case.

Theorem: Agnostic PAC Bound for Finite $\mathcal{H}$

Let $\mathcal{H}$ be finite. For any $\varepsilon, \delta > 0$, if

$$m \geq \frac{1}{2\varepsilon^2} \left(\log|\mathcal{H}| + \log\left(\frac{2}{\delta}\right) \right) \quad (17)$$

then with probability at least $1 - \delta$ over $S \sim \mathcal{D}^m$:

$$\forall h \in \mathcal{H} : |R(h) - \hat{R}_S(h)| \leq \varepsilon \quad (18)$$

Applying this uniform-deviation bound with tolerance $\frac{\varepsilon}{2}$ gives the ERM excess-risk guarantee $R(h_S) \leq \inf_{h \in \mathcal{H}} R(h) + \varepsilon$ with sample complexity $m \geq \frac{2}{\varepsilon^2} (\log|\mathcal{H}| + \log(\frac{2}{\delta}))$.

Proof (Guided Discovery):

Finite Hypothesis Sets: Agnostic Bound

Theorem: Agnostic PAC Bound for Finite $\mathcal{H}$

Let $\mathcal{H}$ be finite. For any $\varepsilon, \delta > 0$, if

$$m \geq \frac{1}{2\varepsilon^2} \left(\log|\mathcal{H}| + \log\left(\frac{2}{\delta}\right) \right) \quad (19)$$

then with probability at least $1 - \delta$ over $S \sim \mathcal{D}^m$:

$$\forall h \in \mathcal{H} : |R(h) - \hat{R}_S(h)| \leq \varepsilon \quad (20)$$

Applying this uniform-deviation bound with tolerance $\frac{\varepsilon}{2}$ gives the ERM excess-risk guarantee $R(h_S) \leq \inf_{h \in \mathcal{H}} R(h) + \varepsilon$ with sample complexity $m \geq \frac{2}{\varepsilon^2} (\log|\mathcal{H}| + \log(\frac{2}{\delta}))$.

Proof (Guided Discovery):

Step 1: Fix a single hypothesis.

For fixed $h \in \mathcal{H}$, the empirical risk is:

$$\hat{R}_S(h) = \frac{1}{m} \sum_{i=1}^m \mathbb{1}_{h(x_i) \neq y_i} \quad (21)$$

Each indicator is an i.i.d. random variable in $[0, 1]$ with expectation $R(h)$.

By Hoeffding's inequality:

$$\mathbb{P}[|\hat{R}_S(h) - R(h)| > \varepsilon] \leq 2 \exp(-2m\varepsilon^2) \quad (22)$$

Step 2: Union bound over all hypotheses.

$$\begin{aligned} \mathbb{P}[\exists h \in \mathcal{H} : |R(h) - \hat{R}_S(h)| > \varepsilon] &\leq \sum_{h \in \mathcal{H}} \mathbb{P}[|R(h) - \hat{R}_S(h)| > \varepsilon] \\ &\leq |\mathcal{H}| \cdot 2 \exp(-2m\varepsilon^2) \end{aligned} \quad (23)$$

Step 3: Solve for sample complexity.

Set $|\mathcal{H}| \cdot 2 \exp(-2m\varepsilon^2) \leq \delta$:

$$\log|\mathcal{H}| + \log 2 - 2m\varepsilon^2 \leq \log \delta \quad (24)$$

$$m \geq \frac{1}{2\varepsilon^2} \left(\log|\mathcal{H}| + \log\left(\frac{2}{\delta}\right) \right) \quad (25)$$

Student Challenge

Exercise: Compare this bound to the realizable case bound from Week 2. Why does the agnostic case require $O(\frac{1}{\varepsilon^2})$ samples instead of $O(\frac{1}{\varepsilon})$?

Connection

From Week 2: In the realizable case, we had $m \geq \frac{1}{\varepsilon} (\log|\mathcal{H}| + \log(\frac{1}{\delta}))$ with $\frac{1}{\varepsilon}$ dependence. The agnostic case requires $\frac{1}{\varepsilon^2}$. For a detailed explanation of why agnostic learning needs the stronger $\frac{1}{\varepsilon^2}$ rate, see the key-takeaway box in the Fundamental Theorem section below.

✓ Checkpoint

Quick check: For $|\mathcal{H}| = 1000$, $\varepsilon = 0.1$, $\delta = 0.05$, how many samples does the agnostic bound require?

Answer: $m \geq \frac{1}{2 \cdot 0.01} (\log 1000 + \log 40) \approx 50(6.9 + 3.7) = 530$ samples. Compare to realizable: $m \geq \frac{1}{0.1} (\log 1000 + \log 20) \approx 99$ samples—over $5\times$ fewer!

week: 3, title: “Section 4: Fundamental Theorem (Regenerated)”)

The Fundamental Theorem of Statistical Learning

We've now seen that uniform convergence makes ERM succeed. But this raises deeper questions:

- **When** does uniform convergence hold?
- Is there a **simple condition** that characterizes learnability?
- What **exactly** controls sample complexity?

The Fundamental Theorem answers all three questions with a single condition: **finite VC dimension**. But before we state it, let's understand why this unification is surprising and powerful.

The Question: How Do These Concepts Relate?

So far in this week, we've encountered three different properties:

Property	Domain	What It Tells Us
Uniform convergence	Statistical	Empirical risk $\hat{R}_S(h)$ approximates true risk $R(h)$ for all $h \in \mathcal{H}$ simultaneously
PAC learnability	Algorithmic	Some algorithm succeeds with polynomial samples
VC dimension $< \infty$	Combinatorial	Hypothesis class can't shatter arbitrarily large sets

🧐 Student Question

“These seem like completely different concepts from different domains. Statistical convergence, algorithmic success, and combinatorial properties—why would they be related? Are they just independent properties that happen to coincide for nice hypothesis classes?”

This confusion is completely natural. We have:

- A **statistical** property (uniform convergence) about empirical estimates
- An **algorithmic** property (PAC learning) about what algorithms can achieve
- A **combinatorial** property (VC dimension) about labeling patterns

Why should a combinatorial counting problem tell us anything about statistical convergence or algorithmic learnability?

The Natural But Wrong Intuition

You might think: “These are separate properties that we study independently. Maybe they're correlated—classes with finite VC dimension tend to be learnable—but they're fundamentally different concepts.”

This seems reasonable because:

- Uniform convergence is about probability and concentration
- PAC learnability is about algorithm design and sample complexity

- VC dimension is about combinatorics and geometry

Each seems to require its own analysis, its own proof techniques, its own characterization.

If they were independent, we'd expect:

- Some classes with finite VC dimension but no uniform convergence
- Some classes that are PAC learnable but have infinite VC dimension
- Some classes with uniform convergence but that aren't algorithmically learnable

This would mean learning theory is a collection of ad-hoc results, each concept needing separate treatment.

The Unification Crisis

Let's see why treating them independently leads to fragmentation:

⚠ Crisis

The Fragmentation Problem

Without a unifying principle, we face:

- For each new hypothesis class, must we check all three properties separately?
- Are there classes where some properties hold but others don't?
- Do different classes require different proof techniques?
- Is there no single answer to "when is learning possible?"

Consider what we'd need to prove for each new hypothesis class $\mathcal{H}$:

Question	Required Analysis	Difficulty
Does UC hold?	Prove concentration for all $h \in \mathcal{H}$ simultaneously	Statistical analysis
Is it PAC learnable?	Design algorithm + prove sample complexity	Algorithm design
Is VCdim finite?	Find largest shatterable set	Combinatorial analysis

The inefficiency: For neural networks, decision trees, kernel methods, etc., we'd need to:

1. Compute VC dimension using combinatorics
2. Separately prove uniform convergence using probability
3. Separately prove PAC learnability using algorithm analysis

Three separate proofs for every new hypothesis class!

The deeper problem: Without a unifying characterization, learning theory feels like a grab-bag of techniques rather than a coherent framework. We have tools but no systematic understanding.

The Surprising Discovery

💡 Aha Moment

The Unification

These three properties are not just correlated—they're **EQUIVALENT**!

Finite VC dimension $\iff$ Uniform convergence $\iff$ PAC learnability

Having any one property **automatically guarantees** the other two.

This is shocking when you first encounter it:

- A **combinatorial** condition (VC dimension) completely determines **statistical** behavior (convergence)
- A **geometric** property (shattering sets) completely determines **algorithmic** feasibility (learning)
- Counting label patterns tells you everything about sample complexity!

Why This Is Powerful

🎯 Key Takeaway

Single Characterization of Learnability

To determine if a machine learning problem is solvable:

1. Identify your hypothesis class $\mathcal{H}$
2. Compute $d = \text{VCdim}(\mathcal{H})$ (combinatorial problem)
3. Decision:
 - If $d < \infty$: **Learnable** with $m = O\left(\frac{d}{\epsilon^2}\right)$ samples
 - If $d = \infty$: **Not learnable** (No Free Lunch)

That's it! One combinatorial calculation answers every question about learnability.

Why this unification matters:

Before the Fundamental Theorem:

- Question: “Can we learn with decision trees?”
- Answer: “Let me check uniform convergence... design an algorithm... prove sample complexity... analyze VC dimension...”
- Result: Four separate analyses

After the Fundamental Theorem:

- Question: “Can we learn with decision trees?”
- Answer: “What's the VC dimension?”
- Calculation: $\text{VCdim}(\text{decision trees of depth } k) = O(k \cdot 2^k)$
- Result: Finite $\rightarrow$ learnable with $m = O\left(k \cdot \frac{2^k}{\epsilon^2}\right)$ samples. Done!

The theoretical beauty: Learning theory is **not** a collection of ad-hoc results. It has a **complete characterization**—a single condition that unifies everything.

The Formal Statement

Now we can appreciate the full statement:

Theorem: Fundamental Theorem of Statistical Learning

In the standard distribution-free binary 0-1 classification setting, for a suitably measurable hypothesis class $\mathcal{H}$ of functions from $\mathcal{X}$ to $\{0, 1\}$, the following are equivalent (each implies all the others):

1. $\mathcal{H}$ has the **uniform convergence property** (empirical risk converges to true risk uniformly for all $h \in \mathcal{H}$)
2. Any ERM rule is a **successful agnostic PAC learner** for $\mathcal{H}$
3. $\mathcal{H}$ is **agnostic PAC-learnable** (learnable without assuming a perfect hypothesis exists)
4. $\mathcal{H}$ is **PAC learnable**
5. Any ERM rule is a **successful PAC learner** for $\mathcal{H}$
6. $\mathcal{H}$ has **finite VC dimension** ($\text{VCdim}(\mathcal{H}) < \infty$)

Moreover, if $\text{VCdim}(\mathcal{H}) = d < \infty$, agnostic PAC learning has sample complexity roughly

$$m = O\left(\frac{d + \log\left(\frac{1}{\delta}\right)}{\varepsilon^2}\right) \quad (26)$$

up to logarithmic factors depending on the theorem version, while realizable PAC learning has a faster $\frac{1}{\varepsilon}$ dependence (instead of $\frac{1}{\varepsilon^2}$).

In words: Within binary classification, having finite VC dimension is both necessary and sufficient for learning. If VC dimension is finite, the class is learnable (both in agnostic and realizable settings); if infinite, it's not learnable. As the theorem title suggests, this is the fundamental characterization of statistical learning.

Understanding the Equivalences

Let's unpack what "equivalent" means:

Definition: Logical Equivalence

Six statements are **equivalent** if each implies all the others. This means:

- If **any one** holds, **all six** hold
- If **any one** fails, **all six** fail
- They're different ways of expressing the **same underlying property**

Think of them as six different "languages" describing the same phenomenon:

- Statement 1: The statistical language (convergence)
- Statement 2-5: The algorithmic language (learning)
- Statement 6: The combinatorial language (VC dimension)

Proof structure: The theorem establishes a cycle:

$$(6) \Rightarrow (1) \Rightarrow (2) \Rightarrow (3) \Rightarrow (4) \Rightarrow (5) \Rightarrow (6) \quad (27)$$

Each implication:

- **(6) $\Rightarrow$ (1):** Combinatorics controls statistics (via Sauer's lemma + symmetrization)
- **(1) $\Rightarrow$ (2):** Statistical reliability enables ERM to succeed (our earlier proof)
- **(2) $\Rightarrow$ (3):** ERM is one particular algorithm
- **(3) $\Rightarrow$ (4):** Agnostic includes realizable as special case
- **(4) $\Rightarrow$ (5):** If learnable, ERM must work (uniqueness of sample complexity)
- **(5) $\Rightarrow$ (6):** Necessity via No Free Lunch (infinite VC $\Rightarrow$ not learnable)

Theorem Landscape

The Equivalence Chain

```

Finite VC dimension (combinatorial)
  ↓ (Sauer + symmetrization)
Uniform convergence (statistical)
  ↓ (our Section 2 proof)
ERM succeeds (algorithmic)
  ↓ (special cases)
PAC learnable (general)
  ↓ (No Free Lunch contrapositive)
Finite VC dimension
  
```

The cycle is complete: starting from any property, we can prove all others.

Why $\frac{1}{\epsilon^2}$ in Agnostic vs $\frac{1}{\epsilon}$ in Realizable?

The theorem gives different rates for agnostic vs realizable learning. Why?

🎯 Key Takeaway

Detection vs Estimation

The different sample complexity rates reflect fundamentally different problems:

Realizable case ($\frac{1}{\varepsilon}$ rate):

- We know some $h^* \in \mathcal{H}$ has zero error: $R(h^*) = 0$
- Task: **Detect** if $R(h) > \varepsilon$ (one-sided test)
- Hoeffding for detection: $\mathbb{P}[\hat{R}_S(h) \leq \frac{\varepsilon}{2} \mid R(h) \geq \varepsilon] \leq \exp(-m \frac{\varepsilon^2}{2})$
- Need $m = O(\frac{1}{\varepsilon})$ samples to reliably detect non-zero error

Agnostic case ($\frac{1}{\varepsilon^2}$ rate):

- No perfect hypothesis: $\min_{h \in \mathcal{H}} R(h)$ may be large
- Task: **Estimate** $R(h)$ within $\pm \varepsilon$ (two-sided estimation)
- Hoeffding for estimation: $\mathbb{P}[|\hat{R}_S(h) - R(h)| > \varepsilon] \leq 2 \exp(-2m \varepsilon^2)$
- Need $m = O(\frac{1}{\varepsilon^2})$ samples for accurate estimation

The fundamental difference: Testing whether something equals zero requires fewer samples than estimating its actual value. This is why agnostic learning (which estimates) is harder than realizable learning (which only tests).

Concrete example:

- To detect if a coin is biased away from 50% by $\varepsilon = 0.1$: $O(100)$ flips suffice
- To estimate the exact bias within ± 0.1 : $O(10000)$ flips needed

Same phenomenon in learning!

Translation Table

🎯 Key Takeaway

Different Perspectives on the Same Property

- **Uniform convergence:** Empirical estimates are reliable for all h simultaneously
- **PAC learnability:** Some algorithm succeeds with polynomial samples
- **VC dimension:** The combinatorial capacity of $\mathcal{H}$

The Fundamental Theorem says: these are not three properties but three **perspectives** on a single underlying phenomenon—bounded effective complexity.

Practical impact:

- **Researcher designing a new model:** Compute VC dimension $\rightarrow$ know immediately if learnable
- **Practitioner choosing sample size:** Use $m = O(\frac{d}{\varepsilon^2})$ formula
- **Theorist proving bounds:** Work in whichever “language” is most convenient (combinatorial, statistical, or algorithmic), knowing all are equivalent

Looking Ahead: The Proof

Progress on backbone: We now know **what** the Fundamental Theorem says—finite VC dimension is the complete characterization. But **how** do we prove it?

The proof has several parts:

1. **Finite VC $\Rightarrow$ Uniform convergence:** This is the hard part, using symmetrization (Section 6)
2. **Uniform convergence $\Rightarrow$ PAC learning:** We already proved this in Section 2!
3. **PAC learning $\Rightarrow$ Finite VC:** This uses the No Free Lunch theorem (Section 8)

The key technical tool is **symmetrization**—a beautiful technique that replaces unknown expectations with empirical estimates on “ghost samples.”

Connection

Why prove all directions? To show equivalence, we must prove both directions:

- **Sufficiency:** Finite VC $\Rightarrow$ learnable (if you have bounded complexity, you can learn)
- **Necessity:** Learnable $\Rightarrow$ finite VC (if you can learn, complexity must be bounded)

Both are essential for a **complete characterization**.

Confidence Check

Self-assessment: Before continuing, ensure you can:

- Explain why uniform convergence is stronger than pointwise
- State the sample complexity: $O\left(\frac{d + \log(\frac{1}{\delta})}{\varepsilon^2}\right)$
- Articulate why the three concepts (statistical, algorithmic, combinatorial) being equivalent is surprising
- Give an example where treating them separately would require three separate proofs

Connection

To Week 4: The Fundamental Theorem tells us **when** learning is possible. Next week: **which** hypothesis class to use when we have choices (model selection, regularization, cross-validation).

Remark

Two-sided vs. one-sided bounds: Uniform convergence naturally gives two-sided bounds: $|R(h) - \hat{R}_S(h)| \leq \varepsilon$ (bounds deviations in both directions). For ERM analysis, we only need the one-sided bound $R(h) \leq \hat{R}_S(h) + \varepsilon$ (bounding how much true risk can exceed empirical risk). Both follow from the same uniform convergence property, but the one-sided version is sufficient for proving PAC learnability.

Proof Sketch: The Equivalence Chain

We prove the equivalences in a cycle: $(4) \Rightarrow (1) \Rightarrow (2) \Rightarrow (3) \Rightarrow (4)$.

Part 1: Finite VC dimension implies uniform convergence

Productive Struggle

Before reading the proof: How would you prove that finite VC dimension implies uniform convergence? The challenge: $\mathcal{H}$ might be infinite, so we can't use union bounds directly. But VC dimension controls the growth function...

Hint: The key insight is **symmetrization**—replace the unknown true risk with empirical risk on an independent “ghost sample.”

Lemma: VC Dimension and Uniform Convergence

If $\text{VCdim}(\mathcal{H}) = d < \infty$, then for any $\varepsilon, \delta > 0$, if

$$m \geq C \frac{d \log(\frac{1}{\varepsilon}) + \log(\frac{1}{\delta})}{\varepsilon^2} \quad (28)$$

for some universal constant C , then with probability $\geq 1 - \delta$:

$$\forall h \in \mathcal{H} : |R(h) - \hat{R}_S(h)| \leq \varepsilon \quad (29)$$

Note on one-sided vs two-sided bounds: This is a two-sided bound (absolute value). For ERM analysis, we typically need only the one-sided bound $R(h) \leq \hat{R}_S(h) + \varepsilon$, but uniform convergence naturally gives the stronger two-sided guarantee. Both hold with the same sample complexity.

Proof idea: The standard proof does not directly union-bound over $\Pi_{\mathcal{H}}(m)$ on the same random sample. Instead:

Step 1: Symmetrize.

Introduce an independent ghost sample S' of size m . Large deviation between $R(h)$ and $\hat{R}_S(h)$ implies, with controlled constants, a large deviation between empirical errors on S and S' .

Step 2: Condition on the combined sample.

Condition on the $2m$ points in $S \cup S'$. On these fixed points, hypotheses matter only through their label patterns.

Step 3: Union bound over effective behaviors.

By Sauer's lemma, the number of distinct dichotomies on the combined sample is at most

$$\Pi_{\mathcal{H}}(2m) \leq \left(\frac{2em}{d} \right)^d \quad (30)$$

for $1 \leq d \leq 2m$.

Step 4: Apply concentration and solve for m .

A union bound over these label patterns, followed by Hoeffding-type concentration, gives

$$m \geq C \frac{d \log(\frac{1}{\varepsilon}) + \log(\frac{1}{\delta})}{\varepsilon^2} \quad (31)$$

for a universal constant C .

Pause and Think

Question: Why does the proof use a ghost sample of size m and then $\Pi_{\mathcal{H}}(2m)$?

Hint: The unknown true risk is replaced by empirical behavior on an independent sample. After conditioning, both samples together contain $2m$ points.

Answer:

We only need to count distinct label patterns on those $2m$ points, which is controlled by Sauer's lemma.

Connection

From Week 2: Sauer's lemma states $\Pi_{\mathcal{H}}(m) \leq \left(\frac{em}{d}\right)^d$ when $\text{VCdim}(\mathcal{H}) = d < \infty$. This polynomial bound (not exponential 2^m) makes union bounds work for infinite classes. The growth function counts distinct label patterns on m points—exactly what we need here!

Scaffolded Derivation (7 steps)

Step-by-step: Why $\Pi_{\mathcal{H}}(2m)$ appears

1. Start: $\mathbb{P}_S \left[\sup_{h \in \mathcal{H}} |R(h) - \hat{R}_S(h)| > \varepsilon \right]$
2. Introduce ghost S' : Replace $R(h)$ with $\frac{1}{m} \sum_{i=1}^m \mathbb{1}_{h(z'_i) \neq y'_i}$
3. Now comparing two empirical quantities: $|\hat{R}_S(h) - \hat{R}_{S'}(h)|$
4. Condition on combined sample $S \cup S'$ (total $2m$ points)
5. On these $2m$ points, distinct hypotheses $\leq \Pi_{\mathcal{H}}(2m)$
6. Union bound over $\Pi_{\mathcal{H}}(2m) \leq \left(\frac{2em}{d}\right)^d$ patterns
7. Result: $m = O\left(\frac{d \log(\frac{1}{\varepsilon}) + \log(\frac{1}{\delta})}{\varepsilon^2}\right)$

Part 2: Uniform convergence implies agnostic PAC learnability

This follows directly from our earlier proof that uniform convergence implies ERM succeeds.

Part 3: Agnostic PAC learnability implies realizable PAC learnability

Trivial: the realizable case is a special case of the agnostic case (when $\min_{h \in \mathcal{H}} R(h) = 0$).

Part 4: Realizable PAC learnability implies finite VC dimension

Productive Struggle

The No Free Lunch construction: To prove this, show that infinite VC dimension makes learning impossible. If $\text{VCdim}(\mathcal{H}) = \infty$, then for any sample size m , there exists a set of size $2m$ that $\mathcal{H}$ shatters. We can construct a distribution where the learner sees only half the points, and labels on the unseen half are random and independent of what the learner observed. Why does this make learning impossible?

Lemma: No Free Lunch for Infinite VC Dimension

If $\text{VCdim}(\mathcal{H}) = \infty$, then $\mathcal{H}$ is not PAC-learnable.

Proof (by contradiction):

Assume $\text{VCdim}(\mathcal{H}) = \infty$ but $\mathcal{H}$ is PAC-learnable by algorithm A .

Since VC dimension is infinite, for any m , there exists a set $C = \{x_1, \dots, x_{2m}\}$ of size $2m$ that is shattered by $\mathcal{H}$.

Consider the following adversarial strategy:

1. Nature chooses distribution $\mathcal{D}$ uniform over C
2. Nature chooses a random labeling $c : C \rightarrow \{0, 1\}$
3. Algorithm A receives sample S of size m from $\mathcal{D}$ labeled by c

Pause and Think

Question: How many points from C are likely to remain unseen?

Hint: A fixed point is unseen with probability $(1 - \frac{1}{2m})^m \approx e^{-\frac{1}{2}}$.

Answer:

So a constant fraction of the $2m$ points remain unseen in expectation. On those unseen points, the random labels are independent of the sample the algorithm observed.

Key observation: Since C is shattered, for any labeling of the m unseen points, there exists $h \in \mathcal{H}$ consistent with the entire sample S and that labeling.

Why this matters: Consider what happens when the learner sees sample S drawn from the $2m$ points:

- The learner observes roughly m labeled points from C (those that fell in S)
- Roughly m points from C remain unseen (with labels determined by the random target)
- On the seen points, the learner can find a hypothesis with zero empirical error (since C is shattered)
- But there are 2^m possible labelings of the unseen m points, and each is equally likely given what the learner observed

The contradiction: Fix any hypothesis h the learner might output. Since all 2^m labelings of the unseen points are consistent with the training data:

- For half of these labelings (chosen uniformly at random), hypothesis h will make errors on at least half the unseen points
- The unseen points constitute a constant fraction ($\approx e^{-\frac{1}{2}} \approx 0.37$) of all $2m$ points
- Therefore, h has expected true error of at least $0.25 \cdot 0.37 > 0.09$ on a constant fraction of the possible target concepts

This holds for **any** hypothesis the learner outputs, regardless of the training sample size m . But PAC learnability requires that for any $\varepsilon > 0$, we can make $R(A(S)) \leq \varepsilon$ with high probability by choosing m large enough. This is impossible when the expected error is bounded below by a constant. Contradiction!

Connection

The No Free Lunch Theorem: This proof shows that without restrictions on $\mathcal{H}$, learning is impossible. Infinite VC dimension means the hypothesis class is “too rich” to learn from finite samples.

Note on rigor: This proof sketch is informal—a fully rigorous proof requires concentration inequalities to bound the number of unseen points and their contribution to the error. See Mohri et al. Theorem 3.21 or Shalev-Shwartz & Ben-David Theorem 6.7 for complete details.

✓ Checkpoint

Quick check: Why doesn't No Free Lunch contradict universal approximators like neural networks?

Answer: Neural networks with fixed architecture have **finite** VC dimension (though very large). No Free Lunch applies only to **infinite** VC dimension. The theorem says you must restrict your hypothesis class—which we do by fixing architecture.

Rademacher Complexity: A Technical Tool

Before diving into symmetrization, we need to introduce **Rademacher complexity**—a key technical tool that measures how well a function class can fit random noise. While VC dimension gives a combinatorial measure of complexity, Rademacher complexity provides a probabilistic measure that's crucial for proving generalization bounds.

Definition: Rademacher Complexity

Let $\mathcal{G}$ be a class of functions $\mathcal{Z} \rightarrow \mathbb{R}$ and $S = \{z_1, \dots, z_m\}$ a fixed sample.

Empirical Rademacher complexity:

$$\hat{\mathcal{R}}_S(\mathcal{G}) = \mathbb{E}_\sigma \left[\sup_{g \in \mathcal{G}} \frac{1}{m} \sum_{i=1}^m \sigma_i g(z_i) \right] \quad (32)$$

where $\sigma = (\sigma_1, \dots, \sigma_m)$ and each σ_i is an independent random variable uniform on $\{-1, +1\}$ (called **Rademacher variables**).

Expected Rademacher complexity:

$$\mathcal{R}_m(\mathcal{G}) = \mathbb{E}_S [\hat{\mathcal{R}}_S(\mathcal{G})] \quad (33)$$

where the expectation is over random samples $S \sim \mathcal{D}^m$.

In words:

- $\hat{\mathcal{R}}_S(\mathcal{G})$ measures how well functions in $\mathcal{G}$ can correlate with random ± 1 noise on a **specific** sample S
- $\mathcal{R}_m(\mathcal{G})$ is the expected correlation over all possible samples of size m
- The Rademacher variables σ_i introduce randomness independent of the data

Intuition: If a function class can fit random noise well (high Rademacher complexity), it's expressive enough to overfit. Low Rademacher complexity means the class is restricted enough to prevent fitting noise—a good property for generalization.

Why “Rademacher”? The random signs $\sigma_i \in \{-1, +1\}$ are called Rademacher random variables (introduced by Hans Rademacher in 1922). They're the simplest non-trivial symmetric random variables.

Example: Rademacher Complexity: Intuition

Consider fitting random labels $\sigma_i \in \{-1, +1\}$ on m points:

- **Finite class:** If $|\mathcal{G}| = N$, then $\mathcal{R}_m(\mathcal{G}) \leq \sqrt{\frac{\log N}{m}}$
- **Linear functions in $\mathbb{R}^d$:** $\mathcal{R}_m(\mathcal{G}) = O\left(\sqrt{\frac{d}{m}}\right)$
- **All functions:** $\mathcal{R}_m(\mathcal{G}) = 1$ (can perfectly correlate with any noise)

Notice: more complex classes have higher Rademacher complexity.

Connection

Relationship to VC dimension: For 0-1 loss classes with $\text{VCdim}(\mathcal{H}) = d$:

$$\mathcal{R}_m(\mathcal{G}) = O\left(\sqrt{\frac{d}{m}}\right) \quad (34)$$

where $\mathcal{G} = \{(x, y) \mapsto \mathbb{1}_{h(x) \neq y} : h \in \mathcal{H}\}$. This connection (via Massart's lemma and Sauer's lemma) is how VC dimension controls generalization through Rademacher complexity.

 Key Takeaway

Rademacher complexity bridges combinatorial complexity (VC dimension) and statistical generalization. It's the technical tool that makes symmetrization work.

week: 3, title: "Section 6: Symmetrization (Regenerated)")

Symmetrization and Ghost Samples: Making the Uncomputable Computable

The Fundamental Theorem tells us that finite VC dimension implies uniform convergence. But **how** do we prove this? The challenge is that $\mathcal{H}$ might be infinite, so we can't use union bounds directly over all hypotheses.

The key technique is **symmetrization**—a beautiful proof method that replaces unknown expectations with computable empirical averages. But before we dive into the technique, let's understand the problem it solves.

The Question: What's Wrong with Direct Analysis?

Recall what we need to prove: if $\text{VCdim}(\mathcal{H}) = d < \infty$, then uniform convergence holds:

$$\mathbb{P}_S \left[\sup_{h \in \mathcal{H}} |R(h) - \hat{R}_S(h)| > \varepsilon \right] \rightarrow 0 \text{ as } m \rightarrow \infty \quad (35)$$

🧐 Student Question

“Why can't we just bound this directly? We have concentration inequalities like Hoeffding. Can't we apply them to each hypothesis and use a union bound over $\Pi_{\mathcal{H}}(m)$ behaviors?”

This seems like a reasonable approach. After all:

- We know Hoeffding's inequality for individual hypotheses
- We know the growth function $\Pi_{\mathcal{H}}(m)$ bounds distinct behaviors
- Union bound should combine them

Let's try this direct approach and see what goes wrong.

The Natural But Wrong Approach: Direct Union Bound

You might think: “Let's bound the deviation for each hypothesis using Hoeffding, then union bound over $\Pi_{\mathcal{H}}(m)$ distinct behaviors.”

Attempt: For a fixed hypothesis h :

$$\mathbb{P} \left[|R(h) - \hat{R}_S(h)| > \varepsilon \right] \leq 2 \exp(-2m\varepsilon^2) \quad (36)$$

If there are $\Pi_{\mathcal{H}}(m)$ distinct behaviors on sample of size m , union bound gives:

$$\mathbb{P} \left[\exists h : |R(h) - \hat{R}_S(h)| > \varepsilon \right] \leq \Pi_{\mathcal{H}}(m) \cdot 2 \exp(-2m\varepsilon^2) \quad (37)$$

With $\Pi_{\mathcal{H}}(m) \leq \left(e \frac{m}{d}\right)^d$ (Sauer's lemma), we need:

$$\left(e \frac{m}{d}\right)^d \cdot 2 \exp(-2m\varepsilon^2) \leq \delta \quad (38)$$

Solving for m :

$$m \geq \frac{1}{2\varepsilon^2} \left(d \log\left(e \frac{m}{d}\right) + \log\left(\frac{2}{\delta}\right) \right) \quad (39)$$

This looks promising! We have m on both sides, but we can solve this...

Pause and Think

Question: Wait—what are we actually bounding here?

Hint: Look carefully at what $R(h)$ is: $R(h) = \mathbb{E}_{z \sim \mathcal{D}}[\ell(h, z)]$

Answer:

This is an **unknown expectation** over the distribution $\mathcal{D}$. We can't compute it!

Where This Breaks: The Uncomputable Expectation Problem

⚠ Crisis

The Fatal Flaw: $R(h)$ is Uncomputable

$$R(h) = \mathbb{E}_{z \sim \mathcal{D}}[\ell(h, z)] = \mathbb{P}_{(x,y) \sim \mathcal{D}}[h(x) \neq y] \quad (40)$$

We don't know the distribution $\mathcal{D}$, so:

- We can't compute $R(h)$ for any specific h
- We can't evaluate whether $|R(h) - \hat{R}_S(h)| > \varepsilon$
- We can't even check if our bound is tight!

Problem: The union bound approach requires bounding deviations of $\hat{R}_S(h)$ from $R(h)$, but $R(h)$ is an unknown quantity involving the true distribution.

The circularity issue:

1. We want to bound: $\mathbb{P}[\sup_h |R(h) - \hat{R}_S(h)| > \varepsilon]$
2. Hoeffding bounds: $\mathbb{P}[|\hat{R}_S(h) - R(h)| > \varepsilon]$ for **fixed** h
3. But $R(h)$ appears in both! It's the unknown quantity we're trying to approximate

Concrete illustration:

- We observe: $\hat{R}_S(h) = 0.15$ on our sample
- We want to know: Is $|R(h) - 0.15| > 0.1$?
- But we need $R(h)$ to answer this!
- $R(h) = \mathbb{E}_{x,y}[\mathbb{1}_{h(x) \neq y}]$ requires knowing $\mathcal{D}$

The dependency problem:

Quantity	What It Is	Can We Compute It?
$\hat{R}_S(h)$	$\frac{1}{m} \sum_{i=1}^m \mathbb{1}_{h(x_i) \neq y_i}$	✓ Yes (from sample S)
$R(h)$	$\mathbb{E}_{\mathcal{D}} [\mathbb{1}_{h(x) \neq y}]$	✗ No (requires knowing $\mathcal{D}$)
$ R(h) - \hat{R}_S(h) $	Deviation of empirical from true	✗ No (need $R(h)$!)

Why we can't just “work around” $R(h)$:

- Can't replace it with another sample's average directly (would need independence structure)
- Can't compute it from $\mathcal{D}$ (we don't know $\mathcal{D}$)
- Can't estimate it from S (that's circular— $\hat{R}_S(h)$ is already our estimate!)

The union bound approach **structurally** requires comparing to an unknown quantity. We need a different approach.

The Key Insight: Replace Unknown with Computable

Aha Moment

The Symmetrization Trick

We can't compute $R(h) = \mathbb{E}_{z \sim \mathcal{D}}[g(z)]$, but we **can** use an independent “ghost sample” S' to create an unbiased estimator:

$$\hat{R}_{S'}(h) = \frac{1}{m} \sum_{i=1}^m g(z'_i) \quad (41)$$

Key property: $\mathbb{E}_{S'}[\hat{R}_{S'}(h)] = R(h)$ — in expectation over ghost samples, the empirical average equals the true risk exactly.

This transforms:

- **Uncomputable:** $|R(h) - \hat{R}_S(h)|$ (involves unknown $R(h)$)
- **Computable (in expectation):** $|\hat{R}_{S'}(h) - \hat{R}_S(h)|$ (both are empirical!)

Why this works:

1. $\mathbb{E}_{S'}[\hat{R}_{S'}(h)] = R(h)$ — the ghost sample average is an unbiased estimator of the true risk (the expectation equals $R(h)$ exactly, not approximately)
2. S' is independent of S (no circular dependency)
3. Both $\hat{R}_S(h)$ and $\hat{R}_{S'}(h)$ are empirical—computable from data
4. We can analyze $|\hat{R}_{S'}(h) - \hat{R}_S(h)|$ combinatorially (it's about labelings on $2m$ points!)
5. The symmetrization proof takes expectation over both S and S' , leveraging the unbiased property to replace $R(h)$ with $\mathbb{E}_{S'}[\hat{R}_{S'}(h)]$

The Ghost Sample Device

Definition: Ghost Sample

Given a sample $S = \{z_1, \dots, z_m\}$ (where each $z_i = (x_i, y_i)$ is a labeled example), a **ghost sample** is an independent sample $S' = \{z'_1, \dots, z'_m\}$ drawn from the same distribution $\mathcal{D}$.

In words: A ghost sample is a second, independent training set of the same size, drawn from the same distribution. We use it as a **theoretical device** to replace unknown expectations with computable empirical averages.

Key property: Both S and S' have the same distribution, so $\mathbb{E}_{S'}[g(z')] = \mathbb{E}_S[g(z)]$ for any function g .

Important: The ghost sample is a proof technique, not something we need in practice! It's a mathematical tool for the analysis, not part of the algorithm.

Wrong Path: But We Don't Have S' in Practice!

Confusion: “You’re using a ghost sample S' in the proof, but when I actually run ERM, I only have one sample S . How does this help?”

Resolution: The ghost sample is a **proof technique**, not an algorithmic requirement:

- **In the proof:** We use S' to replace unknown $R(h)$ with computable $\hat{R}_{S'}(h)$
- **In the algorithm:** We still just use S to compute $\hat{R}_S(h)$ and run ERM
- **Why it works:** The proof shows that the bound holds for the **real** sample S we have

Think of it like mathematical induction: we prove a general statement using an auxiliary construction (like $n + 1$), but we can then apply the result to any specific case.

Visual intuition:

Unknown distribution D :

↓ draw sample
 $\hookrightarrow S = \{z_1, \dots, z_m\}$ (real sample we have)
 $\hookrightarrow S' = \{z'_1, \dots, z'_m\}$ (ghost sample for proof)

Direct approach (fails):

Compare: $\hat{R}_S(h)$ to $R(h) = \mathbb{E}[g(Z)]$
 Problem: Can't compute $R(h)$!

Symmetrization (works):

Compare: $\hat{R}_S(h)$ to $\hat{R}_{S'}(h) = (1/m) \sum g(z'_i)$
 Benefit: Both are empirical-computable!

How Symmetrization Works: The Technique

Key idea: We can't compute $\mathbb{E}[g(z)]$ directly (since we don't know $\mathcal{D}$), but we can approximate it with $\frac{1}{m} \sum_{i=1}^m g(z'_i)$ using an independent ghost sample. This converts a problem involving unknown distributions into one involving only empirical quantities.

Theorem: Symmetrization Lemma (One-Sided Version)**Assumptions:**

- Functions $g : \mathcal{Z} \rightarrow [0, 1]$ are bounded (take values in $[0, 1]$)
- Samples z_i, z'_i are i.i.d. from distribution $\mathcal{D}$

Statement: Let $\mathcal{G}$ be a class of functions $\mathcal{Z} \rightarrow [0, 1]$, and let z_i be i.i.d. from $\mathcal{D}$. Then:

$$\mathbb{E}_S \left[\sup_{g \in \mathcal{G}} \left(\mathbb{E}_Z[g(Z)] - \frac{1}{m} \sum_{i=1}^m g(z_i) \right) \right] \leq 2 \mathbb{E}_{S, S'} \left[\sup_{g \in \mathcal{G}} \frac{1}{m} \sum_{i=1}^m (g(z'_i) - g(z_i)) \right] \quad (42)$$

where $S = \{z_1, \dots, z_m\}$ and $S' = \{z'_1, \dots, z'_m\}$ are independent samples from $\mathcal{D}$.

In words: The expected worst-case **one-sided** deviation between the true expectation and the sample average (left side) is bounded by twice the expected worst-case difference between two independent sample averages (right side).

Key benefit: The right side involves only empirical quantities (comparing two samples), which we can bound using concentration inequalities and combinatorial arguments (like Sauer's lemma).

Important: This is the **one-sided** version (no absolute value on the left). There is also a two-sided version that bounds $\mathbb{E}[\sup | \mathbb{E}[g] - \frac{1}{m} \sum g(z_i) |]$ —interestingly, both versions have the **same factor 2** on the right-hand side, because the factor 2 comes from the triangle inequality $|a - b| \leq |a| + |b|$ applied to split the Rademacher terms (see Step 4), not from handling absolute values on the left. For ERM analysis we only need this one-sided version, which is sufficient to prove PAC learnability.

Tightness note: The one-sided bound may appear loose (why pay factor 2 if we only bound one direction?), but this is unavoidable with the symmetrization technique. More refined analyses using localization or PAC-Bayes can sometimes improve constants, but symmetrization remains the standard approach due to its simplicity and generality.

Breaking down the transformation:

Quantity	Description	Computable?
Left: $\mathbb{E}_Z[g(Z)] - \frac{1}{m} \sum g(z_i)$	Unknown expectation minus sample average	× No
Right: $\frac{1}{m} \sum (g(z'_i) - g(z_i))$	Difference of two empirical averages	✓ Yes!

The price: Factor of 2 and extra randomness (need to account for both S and S').

The benefit: Can now apply combinatorial analysis—on the $2m$ points from $S \cup S'$, there are at most $\Pi_{\mathcal{H}}(2m)$ distinct label patterns!

Why This Solves the Problem

Let's trace through how symmetrization resolves our crisis:

Key Takeaway

The Transformation

Before (uncomputable):

$$\mathbb{P} \left[\sup_{h \in \mathcal{H}} |R(h) - \hat{R}_S(h)| > \varepsilon \right] \quad (43)$$

- Involves unknown $R(h) = \mathbb{E}[\text{loss}]$
- Can't evaluate the deviation
- Union bound doesn't help

After symmetrization (computable):

$$\mathbb{P} \left[\sup_{h \in \mathcal{H}} |\hat{R}_{S'}(h) - \hat{R}_S(h)| > \frac{\varepsilon}{2} \right] \quad (44)$$

- Both $\hat{R}_{S'}(h)$ and $\hat{R}_S(h)$ are empirical
- Deviation is about labelings on $2m$ fixed points
- Can apply Sauer's lemma: at most $\Pi_{\mathcal{H}}(2m) \leq (2e^{\frac{m}{d}})^d$ behaviors
- Union bound over finite (polynomial) number of patterns!

Concrete steps:

1. Replace $R(h)$ with $\hat{R}_{S'}(h)$ (ghost sample average)
2. Now comparing two empirical quantities: $\hat{R}_S(h)$ vs $\hat{R}_{S'}(h)$
3. Condition on the combined $2m$ points in $S \cup S'$
4. On these fixed points, only $\Pi_{\mathcal{H}}(2m)$ distinct hypotheses matter
5. Union bound over $\Pi_{\mathcal{H}}(2m) \leq (2e^{\frac{m}{d}})^d$ (polynomial, not exponential!)
6. Each term bounded by concentration (Hoeffding-type)
7. Result: $m = O(\frac{d}{\varepsilon^2})$ sample complexity

Why the factor of 2 appears:

- We need $|R(h) - \hat{R}_S(h)| \leq \varepsilon$
- Replace $R(h)$ with $\hat{R}_{S'}(h)$: $|\hat{R}_{S'}(h) - \hat{R}_S(h)| \leq \frac{\varepsilon}{2}$
- Account for approximation error: $|R(h) - \hat{R}_{S'}(h)| \leq \frac{\varepsilon}{2}$
- Triangle inequality: $|R(h) - \hat{R}_S(h)| \leq |R(h) - \hat{R}_{S'}(h)| + |\hat{R}_{S'}(h) - \hat{R}_S(h)| \leq \varepsilon$

The factor of 2 is the price for replacing an unknown quantity with a computable one.

Proof Sketch: How Symmetrization Works

Let's see the key steps:

Step 1: Introduce ghost sample

$$\begin{aligned}
\mathbb{E}_S \left[\sup_{g \in \mathcal{G}} \left(\mathbb{E}_Z[g(Z)] - \frac{1}{m} \sum_{i=1}^m g(z_i) \right) \right] &= \mathbb{E}_S \left[\sup_{g \in \mathcal{G}} \left(\mathbb{E}_{z'}[g(z')] - \frac{1}{m} \sum_{i=1}^m g(z_i) \right) \right] \\
&= \mathbb{E}_S \left[\sup_{g \in \mathcal{G}} \mathbb{E}_{S'} \left[\left(\frac{1}{m} \sum_{i=1}^m g(z'_i) \right) - \left(\frac{1}{m} \sum_{i=1}^m g(z_i) \right) \right] \right] \quad (45)
\end{aligned}$$

Key: We've replaced $\mathbb{E}_Z[g(Z)]$ with $\mathbb{E}_{S'}[\frac{1}{m} \sum g(z'_i)]$ (same expectation!).

Step 2: Move supremum outside expectation

By monotonicity of expectation and the inequality $\sup_g \mathbb{E}_{S'}[X_g] \leq \mathbb{E}_{S'}[\sup_g X_g]$:

$$\mathbb{E}_S \left[\sup_{g \in \mathcal{G}} \mathbb{E}_{S'}[\dots] \right] \leq \mathbb{E}_{S,S'} \left[\sup_{g \in \mathcal{G}} \left(\frac{1}{m} \sum_{i=1}^m (g(z'_i) - g(z_i)) \right) \right] \quad (46)$$

Step 3: Use symmetry (Rademacher variables)

Let $\sigma = (\sigma_1, \dots, \sigma_m)$ where σ_i are i.i.d. uniform on $\{-1, +1\}$.

Pause and Think

Question: Why is $(g(z'_i) - g(z_i))$ symmetric to $\sigma_i(g(z'_i) - g(z_i))$?

Hint: Both z_i and z'_i are i.i.d. from $\mathcal{D}$, so swapping them doesn't change distribution.

Answer:

The difference has same distribution as its negative. Multiplying by random sign preserves this.

By symmetry:

$$\mathbb{E}_{S,S'} \left[\sup_g \frac{1}{m} \sum (g(z'_i) - g(z_i)) \right] = \mathbb{E}_{\sigma,S,S'} \left[\sup_g \frac{1}{m} \sum \sigma_i (g(z'_i) - g(z_i)) \right] \quad (47)$$

Step 4: Triangle inequality and identical distributions

The key is to bound the supremum of a sum. For any fixed σ, S, S' , let g^* be the function achieving the supremum. Then:

$$\sup_{g \in \mathcal{G}} \frac{1}{m} \sum_{i=1}^m \sigma_i (g(z'_i) - g(z_i)) = \frac{1}{m} \sum_{i=1}^m \sigma_i (g^*(z'_i) - g^*(z_i)) \quad (48)$$

Now apply the triangle inequality for absolute values. Note that:

$$\begin{aligned}
|\sigma_i (g^*(z'_i) - g^*(z_i))| &= |\sigma_i g^*(z'_i) - \sigma_i g^*(z_i)| \\
&\leq |\sigma_i g^*(z'_i) - 0| + |0 - \sigma_i g^*(z_i)| \\
&= |\sigma_i g^*(z'_i)| + |\sigma_i g^*(z_i)|
\end{aligned} \quad (49)$$

Here, $c = 0$ is the intermediate point in the triangle inequality $|a - b| \leq |a - c| + |c - b|$.

Summing over i and taking supremum:

$$\begin{aligned}
\sup_{g \in \mathcal{G}} \left| \frac{1}{m} \sum_{i=1}^m \sigma_i (g(z'_i) - g(z_i)) \right| &\leq \sup_{g \in \mathcal{G}} \frac{1}{m} \sum_{i=1}^m |\sigma_i (g(z'_i) - g(z_i))| \\
&\leq \sup_{g \in \mathcal{G}} \frac{1}{m} \sum_{i=1}^m (|\sigma_i g(z'_i)| + |\sigma_i g(z_i)|) \\
&= \sup_{g \in \mathcal{G}} \frac{1}{m} \sum_{i=1}^m |\sigma_i g(z'_i)| + \sup_{g \in \mathcal{G}} \frac{1}{m} \sum_{i=1}^m |\sigma_i g(z_i)|
\end{aligned} \tag{50}$$

Since $g : \mathcal{Z} \rightarrow [0, 1]$ are non-negative and $\sigma_i \in \{-1, +1\}$, we have $|\sigma_i g(z)| = g(z)$. Therefore:

$$\begin{aligned}
\ldots &\leq \mathbb{E}_{\sigma, S'} \left[\sup_g \frac{1}{m} \sum \sigma_i g(z'_i) \right] + \mathbb{E}_{\sigma, S} \left[\sup_g \frac{1}{m} \sum \sigma_i g(z_i) \right] \\
&= 2 \mathbb{E}_{\sigma, S} \left[\sup_g \frac{1}{m} \sum \sigma_i g(z_i) \right]
\end{aligned} \tag{51}$$

This final term is $2\hat{\mathcal{R}}_S(\mathcal{G})$, the empirical Rademacher complexity.

Step 5: Connect to VC dimension

By Massart's lemma and Sauer's lemma:

$$\hat{\mathcal{R}}_S(\mathcal{G}) \leq \sqrt{\frac{\log |\mathcal{G}|_S}{m}} \leq \sqrt{\frac{d \log m}{m}} \tag{52}$$

Combining with symmetrization from Step 4:

$$\mathbb{E} \left[\sup_{h \in \mathcal{H}} (R(h) - \hat{R}_S(h)) \right] \leq 2 \sqrt{\frac{d \log(\frac{em}{d})}{m}} \tag{53}$$

This gives us an **expectation** bound. But PAC learning requires **high-probability** guarantees!

Step 6: Concentration via McDiarmid's Inequality

To convert from expectation to high-probability, we use McDiarmid's inequality. The key observation: changing a single sample point z_i changes $\sup_{h \in \mathcal{H}} (R(h) - \hat{R}_S(h))$ by at most $O(\frac{1}{m})$ (bounded differences property).

McDiarmid's inequality then states: with probability $\geq 1 - \delta$,

$$\sup_{h \in \mathcal{H}} (R(h) - \hat{R}_S(h)) \leq \mathbb{E} \left[\sup_{h \in \mathcal{H}} (R(h) - \hat{R}_S(h)) \right] + \sqrt{\frac{\log(\frac{1}{\delta})}{2m}} \tag{54}$$

Substituting our expectation bound from Step 5:

$$\sup_{h \in \mathcal{H}} (R(h) - \hat{R}_S(h)) \leq 2 \sqrt{\frac{d \log(\frac{em}{d})}{m}} + \sqrt{\frac{\log(\frac{1}{\delta})}{2m}} \tag{55}$$

This is the final high-probability generalization bound. The second term $\sqrt{\frac{\log(\frac{1}{\delta})}{2m}}$ is the **confidence term** that comes from McDiarmid—it ensures the bound holds with probability $\geq 1 - \delta$.

Result: Sample complexity $m = O(\frac{d}{\epsilon^2})$ for agnostic PAC learning.

Key Takeaway

Symmetrization converts a problem involving unknown expectations into one involving only empirical quantities. The price is a factor of 2 and introduction of Rademacher variables. The benefit is that we can now use combinatorial arguments (Sauer's lemma) to bound the complexity.

Wrong Path: Common Confusion: Where Does the Factor 2 Go?

The Question: “The Symmetrization Lemma has a factor of 2. But when we apply it to get the generalization bound (next theorem), we still have factor 2 in front of the square root. Shouldn't it be 4 after combining constants?”

The Answer: The factor 2 in the Symmetrization Lemma is **correctly preserved** in the final bound! Here's why:

The Full Chain:

1. Symmetrization: $\mathbb{E}[\sup(R(h) - \hat{R}_S(h))] \leq 2 \cdot \hat{\mathcal{R}}_m(\mathcal{H}) \quad \leftarrow \text{factor } \mathbf{2}$
2. Massart's Lemma: $\hat{\mathcal{R}}_m(\mathcal{H}) \leq \sqrt{\frac{\log|\mathcal{H}|_S}{m}} \quad \leftarrow \text{no extra factor}$
3. Sauer's Lemma: $\log|\mathcal{H}|_S \leq d \log(\frac{em}{d}) \quad \leftarrow \text{no extra factor}$
4. Combined: $\mathbb{E}[\sup(R(h) - \hat{R}_S(h))] \leq 2\sqrt{\frac{d \log(em/d)}{m}} \quad \leftarrow \text{factor } \mathbf{2} \text{ preserved}$
5. McDiarmid: Add $\sqrt{\frac{\log(2/\delta)}{2m}}$ for high-probability bound

Final bound: $R(h) \leq \hat{R}_S(h) + 2\sqrt{\frac{2d \log(2em/d)}{m}} + \sqrt{\frac{\log(2/\delta)}{2m}}$

The factor **2** in front comes directly from symmetrization step 1. The factor **2** inside the log comes from Sauer's lemma refinements. These are different 2's!

Key insight: We're bounding $\sup(R(h) - \hat{R}_S(h))$ (one-sided, no absolute value), which is exactly what ERM needs. The factor 2 from symmetrization is the **only** factor 2 in the main term, and it appears explicitly in the final bound.

Remark**Why does one-sided symmetrization still have factor 2?**

You might wonder: “If we only bound one direction (no absolute value), why do we pay the same factor of 2 as the two-sided version?”

Answer: The factor 2 comes from the **triangle inequality** applied to Rademacher terms in Step 4:

$$\sigma_i(g(z'_i) - g(z_i)) = \sigma_i g(z'_i) - \sigma_i g(z_i) \quad (56)$$

We bound:

$$|\sigma_i g(z'_i) - \sigma_i g(z_i)| \leq |\sigma_i g(z'_i)| + |\sigma_i g(z_i)| \quad (57)$$

This splitting introduces factor 2 regardless of whether the original LHS has absolute values—it’s a consequence of how we handle the ghost sample difference.

Is this loose? Yes, potentially! The symmetrization technique is not optimal in terms of constants. More refined approaches exist:

- **Localization techniques** can reduce constants in some cases
- **PAC-Bayes bounds** use different proof techniques with better constants
- **Direct Rademacher analysis** without ghost samples can be tighter

However, symmetrization remains the standard pedagogical approach because:

1. Conceptually clear (replace unknown with ghost sample)
2. Works for general function classes
3. Leads directly to VC dimension characterization
4. The factor 2 is absorbed by other constants in practice (the $O(\cdot)$ notation)

The takeaway: Factor 2 is the “tax” we pay for the symmetrization technique, not a fundamental limit of learning theory.

Looking Back: What We Achieved

Let’s appreciate what symmetrization accomplished:

The original challenge:

- Need to prove: finite VC dimension $\Rightarrow$ uniform convergence
- Obstacle: $\mathcal{H}$ might be infinite (can’t union bound over all hypotheses)
- Deeper obstacle: $R(h)$ is uncomputable (unknown distribution)

The symmetrization resolution:

- Replace unknown $R(h)$ with computable $\hat{R}_{S'}(h)$ (ghost sample)
- Transform to empirical problem: compare $\hat{R}_S(h)$ vs $\hat{R}_{S'}(h)$
- Condition on $2m$ fixed points from $S \cup S'$
- Apply Sauer’s lemma: only $\Pi_{\mathcal{H}}(2m) \leq \left(2e \frac{m}{d}\right)^d$ distinct patterns
- Union bound over polynomial (not exponential or infinite!) number of patterns

Theoretical beauty:

- Converts distributional problem to combinatorial problem
- Makes infinite hypothesis class tractable via finite growth function
- Unifies probability (concentration) with combinatorics (VC dimension)

Connection

Connecting to VC dimension via Massart's Lemma: The Rademacher complexity $\hat{\mathcal{R}}_S(\mathcal{G})$ can be bounded using the growth function. Massart's lemma states that for functions taking values in $\{0, 1\}$:

$$\hat{\mathcal{R}}_S(\mathcal{G}) \leq \sqrt{\frac{\log |\mathcal{G}|_S}{m}} \quad (58)$$

where $|\mathcal{G}|_S$ is the number of distinct functions in $\mathcal{G}$ on sample S . By Sauer's lemma, $|\mathcal{G}|_S \leq \Pi_{\mathcal{H}}(m) \leq \left(\frac{em}{d}\right)^d$, giving $\hat{\mathcal{R}}_S(\mathcal{G}) = O\left(\sqrt{\frac{d \log m}{m}}\right)$. This is how VC dimension controls generalization through Rademacher complexity.

Connection

To Week 5: Symmetrization leads to **Rademacher complexity**, a more refined measure than VC dimension. Rademacher gives data-dependent bounds that can be much tighter in practice.

Confidence Check

Self-assessment: Before continuing, ensure you can:

- Explain why $R(h)$ is uncomputable (involves unknown distribution)
- Describe how ghost sample S' provides a computable alternative
- Articulate why this transforms distributional problem to combinatorial one
- State the price (factor of 2) and benefit (can apply Sauer's lemma) of symmetrization

Use in learning theory: Applied to the supremum $\sup_{h \in \mathcal{H}} |R(h) - \hat{R}_S(h)|$, which changes by at most $O\left(\frac{1}{m}\right)$ when one sample point changes. This gives the high-probability bound from the expectation bound.

Student Challenge

Exercise: Work through the full proof using the symmetrization lemma and Sauer's lemma. This is a key technique you'll use repeatedly.

 Synthesis Challenge

Compare proof techniques: For $\mathcal{H}$ = linear classifiers in $\mathbb{R}^{10}$ with $\text{VCdim} = 11$ and $m = 1000$ samples:

1. Union bound over finite class (if discretized)
2. Union bound over growth function (Sauer's lemma)
3. Rademacher complexity (preview of Week 5)

What bound does each give? When is each most useful? How do they relate to the Fundamental Theorem?

Sample Complexity Bounds

The Fundamental Theorem and symmetrization give us **existence** results: they tell us learning is possible when $\text{VCdim}(\mathcal{H}) < \infty$. But practitioners need concrete numbers: **How many samples do I actually need?**

This section provides the answer: we'll see both upper bounds ($O(\frac{d}{\varepsilon^2})$ samples suffice) and matching lower bounds ($\Omega(\frac{d}{\varepsilon^2})$ required in worst case). These tight bounds tell us that our theory is not just correct—it's **optimal**.

Upper Bounds

Theorem: Sample Complexity Upper Bound

Let $\mathcal{H}$ have $\text{VCdim}(\mathcal{H}) = d < \infty$. For agnostic PAC learning with accuracy ε and confidence δ :

$$m = O\left(\frac{d \log(\frac{1}{\varepsilon}) + \log(\frac{1}{\delta})}{\varepsilon^2}\right) \quad (59)$$

samples suffice.

This formalizes the sample complexity statement from the Fundamental Theorem (Section 5).

Proof: Follows from the Fundamental Theorem and uniform convergence bound.

Example: Concrete Sample Complexities

For $\varepsilon = 0.1$, $\delta = 0.05$:

- **Intervals on $\mathbb{R}$:** $d = 2$, so $m \approx 2000$
- **Linear classifiers in $\mathbb{R}^{10}$:** $d = 11$, so $m \approx 11000$
- **Linear classifiers in $\mathbb{R}^{100}$:** $d = 101$, so $m \approx 101000$
- **Axis-aligned rectangles in $\mathbb{R}^2$:** $d = 4$, so $m \approx 4000$

Notice the linear dependence on d !

Lower Bounds

Theorem: Sample Complexity Lower Bounds

Let $\mathcal{H}$ have $\text{VCdim}(\mathcal{H}) = d \geq 2$. In the realizable setting, some distributions require

$$m = \Omega\left(\frac{d}{\varepsilon} + \frac{\log(\frac{1}{\delta})}{\varepsilon}\right) \quad (60)$$

samples. In the agnostic setting, some distributions require

$$m = \Omega\left(\frac{d}{\varepsilon^2} + \frac{\log(\frac{1}{\delta})}{\varepsilon^2}\right) \quad (61)$$

samples.

Proof technique: Probabilistic method. Construct a hard distribution on a shattered set and show that with fewer samples, no algorithm can distinguish between random labelings.

 Key Takeaway

The upper and lower bounds match up to logarithmic factors! This means our bounds are **tight**—we cannot do fundamentally better.

 Checkpoint

Quick check: Why is the agnostic lower bound $\Omega(\frac{d}{\varepsilon^2})$ while realizable is $\Omega(\frac{d}{\varepsilon})$?

Answer: In realizable, we know some $h \in \mathcal{H}$ has zero error. In agnostic, we must estimate error rates from noisy samples, requiring more data. The $\frac{1}{\varepsilon^2}$ vs. $\frac{1}{\varepsilon}$ reflects estimating a mean (agnostic) vs. testing if mean is zero (realizable).

 Construct Your Own

Design a hard instance: For $\mathcal{H}$ = intervals on $\mathbb{R}$ with $\text{VCdim} = 2$, construct a distribution requiring $\Omega(\frac{2}{\varepsilon^2})$ samples in agnostic case.

Hint: Place probability mass on 3 points, make optimal classifier have error $\frac{1}{3}$. Any classifier from few samples will have error $\gg \frac{1}{3}$.

The No Free Lunch Theorem

So far we've assumed $\text{VCdim}(\mathcal{H}) < \infty$. But what if this doesn't hold?

The No Free Lunch Theorem delivers a stark answer: **No**. Without restricting the hypothesis class, learning is fundamentally impossible. This completes the picture: finite VC dimension isn't just sufficient for learning—it's also **necessary**.

Theorem: No Free Lunch Theorem (Informal)

For any algorithm A and sample size m , one can construct a distribution where A has constant probability of constant error. No learning algorithm works for all possible hypothesis classes.

Key insight: If $\text{VCdim}(\mathcal{H}) = \infty$, an adversary can construct scenarios where the learner sees only half the points, and labels on unseen points are independent of what was observed. This makes learning impossible.

Connection

The theorem says: you **must** restrict your hypothesis class (e.g., finite VC dimension) to make learning possible. Inductive bias is necessary, not optional.

Structural Risk Minimization (Brief Preview)

We've learned when learning is possible (finite VC dimension) and how many samples we need. But how do we choose among multiple hypothesis classes?

Structural Risk Minimization (SRM) provides a theoretical answer: choose the class minimizing training error **plus** a complexity penalty proportional to $\sqrt{\frac{d}{m}}$.

Definition: Structural Risk Minimization

Given nested classes $\mathcal{H}_1 \subseteq \mathcal{H}_2 \subseteq \dots$, SRM chooses:

$$\hat{k} = \arg \min_k \left(\hat{R}_S(h_k) + \sqrt{\frac{\text{VCdim}(\mathcal{H}_k) \log m}{m}} \right) \quad (62)$$

where h_k minimizes training error in $\mathcal{H}_k$.

Interpretation: SRM balances fit (low training error) against complexity (VC dimension). This is the bias-variance tradeoff in action—the theoretical foundation for cross-validation and regularization (Week 5).

Connection

To Week 5: SRM motivates practical model selection via cross-validation and regularization penalties.

Summary and Connections

Key Insights

1. **Agnostic PAC learning** removes the realizability assumption and requires competing with the best hypothesis in $\mathcal{H}$
2. **Uniform convergence** is the key property: empirical risk must approximate true risk uniformly for all hypotheses
3. **The Fundamental Theorem** completely characterizes learnability: finite VC dimension is necessary and sufficient
4. **Symmetrization** is a powerful technique for deriving generalization bounds using ghost samples
5. **No Free Lunch** tells us that restrictions on $\mathcal{H}$ are necessary—learning requires inductive bias
6. **Sample complexity** scales as $O\left(\frac{d}{\varepsilon^2}\right)$ where $d = \text{VCdim}(\mathcal{H})$, and this is tight

Key Takeaway

The Fundamental Theorem of Statistical Learning is the cornerstone of learning theory. It tells us exactly when learning is possible and how many samples we need. Everything else is refinement and application of this core result.

Proof Template: Proving uniform convergence for your class

To show $\mathcal{H}$ satisfies uniform convergence:

1. **Verify finite VC:** Show $\text{VCdim}(\mathcal{H}) = d < \infty$
2. **Apply Sauer:** $\Pi_{\mathcal{H}}(m) \leq \left(\frac{em}{d}\right)^d$
3. **Symmetrization:** Introduce ghost sample, bound $\mathbb{P}\left[\sup_{h \in \mathcal{H}} |R(h) - \hat{R}_S(h)| > \varepsilon\right]$
4. **Union bound:** Over $\Pi_{\mathcal{H}}(2m)$ behaviors
5. **Concentration:** Apply Hoeffding to each
6. **Solve for m :** Get $m = O\left(\frac{d \log(\frac{1}{\varepsilon}) + \log(\frac{1}{\delta})}{\varepsilon^2}\right)$

Proof Template: Proving sample complexity lower bound

To show $\mathcal{H}$ requires $\Omega\left(\frac{d}{\varepsilon^2}\right)$ samples:

1. **Find shattered set:** Identify set C of size d that $\mathcal{H}$ shatters
2. **Construct distribution:** Uniform over C
3. **Random labeling:** Choose target uniformly
4. **Analyze learning:** With $m < c \frac{d}{\varepsilon^2}$, learner sees too few points
5. **Calculate error:** Expected error 0.5 on unseen; cannot achieve $\leq \varepsilon$

Connections to Other Topics

Connection

To Week 4: Bias-variance tradeoff and regularization implement the complexity-fit balance in practice.

To Week 6-7: Boosting and ensemble methods—how does combining weak learners affect complexity?

To Week 11: Deep learning theory. Why do neural networks generalize despite having huge VC dimension?

Connection

Building on Week 2: Last week we computed VC dimensions for specific classes (intervals, rectangles, linear classifiers). This week we saw **why** VC dimension matters: it's the key quantity controlling sample complexity. The classes we analyzed are all learnable precisely because they have finite VC dimension.

Connection

Looking ahead to Week 4: Next week focuses on **practical** learning:

- Cross-validation for model selection (empirical alternative to SRM)
- Regularization (penalizing complexity in objective)
- Bias-variance decomposition (quantifying approximation-estimation trade-off)
- Validation set methods (practical tools for avoiding overfitting)

Mental Model: The Learning Theory Landscape

The chain of learning theory:

Hypothesis Class $\mathcal{H} \rightarrow$ Complexity $\text{VCdim}(\mathcal{H}) = d \rightarrow$ Sample Complexity $m = O\left(\frac{d}{\varepsilon^2}\right) \rightarrow$ Uniform Convergence $\sup_{h \in \mathcal{H}} |R(h) - \hat{R}_S(h)| \leq \varepsilon \rightarrow$ ERM Success $R(h_S) \leq \min_{h \in \mathcal{H}} R(h) + \varepsilon$

Every arrow is an implication in the Fundamental Theorem. The chain holds iff $d < \infty$.

Closing Challenge

Opening Challenge

Revisiting the Opening Question:

Your training data is noisy and your hypothesis class may be misspecified. Can you still learn?

Answer: Yes! The Fundamental Theorem guarantees that if $\text{VCdim}(\mathcal{H}) < \infty$, then:

- ERM will find a hypothesis close to the best in $\mathcal{H}$
- Sample complexity is $O\left(\frac{d}{\epsilon^2}\right)$
- This holds regardless of noise or misspecification

Practical lesson:

- Choose hypothesis classes with finite (and small) VC dimension
- Use enough samples: $m \gg \frac{d}{\epsilon^2}$
- Balance model complexity with sample size
- Validate on held-out data to estimate true risk

Author's Take

The Fundamental Theorem provides a complete answer within the standard binary classification framework. It connects four seemingly different concepts (uniform convergence, PAC learnability, VC dimension) into a single unified framework.

In practice, most modern machine learning operates in the agnostic setting—we rarely have perfect data or perfectly specified models. The Fundamental Theorem tells us this is okay: as long as our hypothesis class has finite VC dimension and we have enough samples, learning is possible.

The key insight is that **complexity matters**. Not all hypothesis classes are learnable, and the sample complexity depends critically on the VC dimension. This motivates the entire field of model selection and regularization, which we'll explore in coming weeks.

Synthesis Challenge

Capstone: Design a complete learning system

Problem: Predict disease from 20 biomarkers with $m = 500$ samples.

Tasks:

1. Propose three hypothesis classes and estimate their VC dimensions
2. Which are learnable with $\epsilon = 0.1$, $\delta = 0.05$?
3. Use SRM with training errors $\text{Remp}_1 = 0.15$, $\text{Remp}_2 = 0.10$, $\text{Remp}_3 = 0.05$
4. How much more data to make the complex class competitive?
5. What inductive biases are implicit in your choices?

This integrates: agnostic PAC, VC dimension, sample complexity, uniform convergence, SRM, No Free Lunch.

Example: Medical diagnosis finale

You started with noisy data and misspecified model (linear for nonlinear problem). Through Week 3's theory:

- Your problem is **agnostic** (best linear has 15% error)
- With $m = 1000$ and $\text{VCdim} = 11$, you can find classifier with 17% error (Fundamental Theorem)
- ERM works because uniform convergence holds (finite VC)
- You needed $m = O(\frac{11}{0.01}) \approx 1100$ samples ✓
- Alternative classes (quadratic) might fit better but need more samples or risk overfitting (SRM)

This is learning theory in action: rigorous guarantees for messy, real-world problems.